

Mutshidzi Madzivhandila

Computer Scientist, Software Developer

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Professional Summary

As a final-year BSc student in Applied Mathematics and Computer Science, I have honed my skills as a proficient Python and JavaScript developer, particularly adept at building both academic and independent technical projects. My experience as a Computer Science Intern/Test Automation Developer at Stellenbosch University has equipped me with the ability to develop and maintain test automation scripts, significantly reducing manual processing time and enhancing workflow efficiency. I have a strong foundation in integrating APIs and AI tools to automate complex processes, which aligns well with the requirements of a Web Developer role. My work in sales has further refined my communication and problem-solving skills, essential for collaborating effectively within a team. I possess a keen eye for detail, especially in testing and optimizing software systems, ensuring high-quality code and improved system performance. My project portfolio, including a full-stack SaaS platform and an epidemiological disease spread simulator, demonstrates my ability to apply technical skills to real-world challenges. These projects highlight my proficiency in React, Flask, and other relevant technologies, as well as my ability to design and implement scalable and maintainable codebases. I am eager to bring my technical expertise, creativity, and enthusiasm to a junior web developer position, where I can contribute to innovative financial service solutions while continuing to grow professionally.

Skills	
Your Skills	
Python Python is a versatile programming language used for web development, making it essential for building dynamic web applications.	Java Java is a robust, object-oriented programming language that is crucial for developing scalable web applications.
JavaScript JavaScript is a core technology for web development, enabling interactive and dynamic content on web pages.	Object-Oriented Programming Object-Oriented Programming is a paradigm that helps in organizing code efficiently, crucial for developing maintainable web applications.
Algorithms & Data Structures Understanding algorithms and data structures is vital for optimizing web application performance and efficiency.	Data Analysis Data analysis skills are important for interpreting user data to improve web application functionality and user experience.
Mathematical Modelling Mathematical modelling aids in creating algorithms that can enhance the functionality of financial service applications.	Problem Solving Problem-solving skills are essential for identifying and fixing issues in web development projects.
Analytical Thinking Analytical thinking is crucial for breaking down complex problems and developing effective web solutions.	Git Git is a version control system that is essential for tracking changes and collaborating on web development projects.
GitHub / GitLab GitHub and GitLab are platforms for hosting and managing code repositories, facilitating team collaboration in web development.	Google Apps Script Google Apps Script can automate tasks and integrate Google services, enhancing productivity in web projects.
VS Code VS Code is a popular code editor that supports various programming languages, making it ideal for web development.	LaTeX LaTeX is used for creating technical documentation, which is important for maintaining clear project records.
Software Development Software development skills are fundamental for building and maintaining web applications.	Workflow Automation Workflow automation helps streamline development processes, increasing efficiency in web projects.
API Integration API integration is crucial for connecting web applications with external services and enhancing functionality.	Scripting Scripting skills are important for automating repetitive tasks and improving web application efficiency.
Debugging Debugging is essential for identifying and fixing errors in code, ensuring web applications run smoothly.	HTML HTML is the standard markup language for creating web pages, forming the backbone of web development.
CSS CSS is used for styling web pages, making it crucial for creating visually appealing web applications.	React React is a JavaScript library for building user interfaces, essential for developing dynamic and responsive web applications.
Databases & SQL Knowledge of databases and SQL is important for managing and retrieving data in web applications.	Command Line Tools Command line tools are essential for performing various development tasks efficiently in a web development environment.
Technical Documentation Technical documentation is crucial for maintaining clear and comprehensive records of web development processes.	Communication Effective communication skills are vital for collaborating with team members and stakeholders in web development projects.
Collaboration Collaboration skills are important for working effectively in a team to achieve web development goals.	Time Management Time management is crucial for meeting deadlines and ensuring timely delivery of web development projects.
Adaptability Adaptability is essential for keeping up with evolving technologies and methodologies in web development.	
Tools and Platforms	
Jira Jira is a project management tool used for tracking tasks, bugs, and project progress. This job requires Jira to efficiently manage and track the development tasks and ensure that projects are completed on time and within scope.	Bitbucket Bitbucket is a Git repository management solution designed for professional teams. This job needs Bitbucket for version control and collaboration on code, ensuring that multiple developers can work on the same project without conflicts.
Confluence Confluence is a collaboration tool used to help teams collaborate and share knowledge efficiently. This job requires Confluence to document processes, share information, and facilitate team communication.	Amazon Web Services (AWS) AWS is a comprehensive cloud computing platform provided by Amazon. This job could benefit from AWS knowledge to deploy, manage, and scale applications in a cloud environment, ensuring high availability and performance.

Frameworks

Vue.js Vue.js is a progressive JavaScript framework used for building user interfaces. This job requires Vue.js to develop dynamic and responsive web applications, enhancing the user experience and interface design.	jQuery jQuery is a fast, small, and feature-rich JavaScript library. This job requires jQuery for simplifying HTML document traversing, event handling, and animation, making it easier to create interactive web pages.
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Data Skills

JSON JSON (JavaScript Object Notation) is a lightweight data-interchange format. This job requires JSON for data exchange between client and server, ensuring efficient communication and data handling in web applications.	
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Soft Skills

Attention to Detail Attention to detail involves the ability to achieve thoroughness and accuracy when accomplishing a task. This job requires excellent attention to detail to ensure that all HTML and CSS code is tested and functions correctly across different browsers.	Flexibility Flexibility is the ability to adapt to new conditions and changes. This job requires flexibility to learn new technologies and respond to changing project requirements, ensuring continuous professional growth and project success.
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Additional Skills

Responsive Web Design Responsive web design is an approach to web design that makes web pages render well on a variety of devices and window or screen sizes. This job requires it to ensure that all HTML is device responsive for optimal end-user experience.	Cross-Browser Compatibility Cross-browser compatibility ensures that a website or web application works across different web browsers. This job requires a strong understanding of cross-browser compatibility to ensure consistent functionality and appearance.
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Experience	
COMPUTER SCIENCE INTERN /TEST AUTOMATION DEVELOPER 2021 – Present	
Stellenbosch University • Stellenbosch, South Africa	
→ Developed and maintained test automation scripts using Python, JavaScript, and Google Apps Script, reducing manual processing time and improving workflow efficiency. → Built and optimized automation tools and document-processing pipelines to extract, clean, and format large datasets for academic and software development projects. → Integrated APIs and AI tools into existing systems to automate data transformation, reporting, and content generation processes. → Collaborated with developers to debug, test, and optimize software systems, improving code quality, automation coverage, and overall system performance. → Contributed to the development of UI/UX components for internal interfaces, enhancing user experience and interface functionality. → Assisted in the design and execution of digital marketing materials, aligning with brand objectives and marketing strategies. → Participated in Agile development processes, ensuring timely delivery of high-quality software solutions.	
SALES REPRESENTATIVE 2023 – 2025	
M AND L HPC BRANDS GROUP (PTY) LTP • Johannesburg, South Africa	
→ Delivered remote customer engagement and product support, strengthening communication, problem-solving, and client relationship management skills. → Adapted messaging and sales approaches to interact effectively with diverse customers, improving customer satisfaction and engagement. → Managed customer inquiries and product information requests, demonstrating patience, active listening, and clear explanation of complex information. → Maintained consistent remote collaboration with team members and management, supporting sales targets and customer service objectives. → Applied analytical skills to assess customer needs and recommend suitable products, contributing to increased sales and customer loyalty. → Utilized CRM tools to track customer interactions and feedback, providing insights for product improvement and service enhancement. → Participated in team meetings and training sessions to stay updated on product knowledge and sales techniques.	

Education
Bachelor of Science: Applied Mathematics
Stellenbosch University • Stellenbosch, South Africa
December 2026
Minimum Average: 70
Relevant Coursework:
→ Algorithms & Data Structures: This course covers essential algorithms and data structures, crucial for developing efficient and scalable web applications. → Web Programming & Databases: Focuses on web development and database management, directly applicable to creating dynamic and responsive web applications. → Operations Research: Teaches optimization techniques, relevant for improving application performance and resource management. → Linear Algebra: Provides foundational knowledge in mathematical modeling, essential for data analysis and algorithm development. → Analysis: Covers mathematical analysis techniques, applicable to performance evaluation and optimization of web applications. → Mathematical Modelling: Involves creating mathematical representations of real-world systems, useful for developing complex web solutions. → WA for Aurora Netball Team, 2021 to 2025: Demonstrates teamwork and leadership skills, valuable for collaborative web development projects. → Sales Representative Apprenticeship, M AND L HPC BRANDS GROUP (PTY) LTP, Completed 2025: Highlights communication and customer service skills, essential for client-facing roles in web development.

Project Experience

FrontendJobs AI-Powered Job Search Platform

React, Vite, Flask, PostgreSQL, OpenAI GPT-4o, Python, DigitalOcean

- Architected and deployed a full-stack SaaS platform using React (Vite) and Flask, integrating OpenAI GPT-4o to generate ATS-optimised CVs and cover letters tailored to specific job descriptions.
- Engineered a semantic similarity scoring engine using OpenAI text embeddings and cosine similarity, delivering real-time ATS match scores, skills gap analysis, and missing keyword detection with weighted composite scoring.
- Designed and implemented RESTful API endpoints with JWT authentication, Google OAuth 2.0, and PostgreSQL persistence, deployed to DigitalOcean with Nginx reverse proxy and SSL via Certbot.
- Built an AI tools suite — including Interview Prep, LinkedIn Bio Optimiser, Salary Estimator, and Skills Gap Report — enabling users to maximise job application success through data-driven career intelligence.
- Demonstrated strong skills in writing custom HTML, CSS, and JavaScript for responsive web design, ensuring optimal user experience across devices.
- Applied Agile methodologies in project management, enhancing collaboration and timely delivery of project milestones.
- Utilized Jira and Bitbucket for project tracking and version control, ensuring seamless integration and deployment processes.
- Showcased ability to integrate with REST APIs, enhancing platform functionality and user engagement.

SEIR/SEIRD Epidemiological Disease Spread Simulator

SciPy, Matplotlib

- Engineered a compartmental epidemiological simulation using SEIR and SEIRD differential equation models to replicate real-world infectious disease spread dynamics across population groups.
- Implemented numerical solvers using SciPy to compute differential equation systems, producing time-series outputs that tracked Susceptible, Exposed, Infected, Recovered, and Deceased compartments.
- Conducted parameter sensitivity analysis by systematically varying transmission rate, recovery rate, and mortality coefficients, generating comparative visualisations using Matplotlib to assess outbreak severity under different scenarios.
- Validated simulation outputs against published epidemiological data, demonstrating the ability to translate complex mathematical theory into accurate, reproducible computational models.
- Enhanced understanding of data visualization techniques, crucial for creating compelling digital marketing materials.
- Showcased problem-solving skills through model validation and optimization, aligning with the technical demands of the role.
- Developed analytical skills in assessing model outputs, applicable to evaluating marketing campaign performance.
- Demonstrated capability to work independently on complex projects, a valuable asset for handling multiple projects simultaneously.

React Task Manager — Component-Based Web Application

React, JavaScript, HTML, CSS, React Router

- Architected a React single-page application using a component-based design pattern, implementing React Router for client-side navigation and dynamic content rendering across multiple views.
- Built real-time search and multi-criteria filtering functionality using React hooks (useState, useEffect), enabling efficient data querying and responsive UI updates without page reloads.
- Applied frontend engineering best practices including reusable component composition, prop drilling management, and separation of concerns to produce a scalable and maintainable codebase.
- Implemented responsive layouts using CSS Flexbox and Grid, ensuring consistent cross-device and cross-browser compatibility across desktop and mobile viewports.
- Displayed strong skills in writing custom HTML and CSS, crucial for developing creative content and UI/UX components.
- Demonstrated proficiency in JavaScript and jQuery, essential for creating interactive and animated web elements.
- Utilized Agile methodologies to manage project tasks and timelines effectively.
- Showcased ability to design intuitive user interfaces, enhancing user experience and engagement.

Algorithm & Data Structures Implementation Library

Python, Java, Data Structures, Big-O Analysis

- Implemented a library of core computer science algorithms in Python and Java, including binary search, quicksort, merge sort, depth-first search, and breadth-first search for practical problem-solving applications.
- Performed rigorous time and space complexity analysis using Big-O notation, benchmarking algorithm performance across varying input sizes to justify data structure selection and optimisation decisions.

- Solved over 30 logic-based programming challenges by decomposing problems into modular, testable components — applying dynamic programming, recursion, and greedy strategies to derive optimal solutions.
- Built systematic debugging and test validation workflows, writing edge-case test suites in Python to verify correctness of all algorithm implementations before deployment.
- Enhanced problem-solving and analytical skills, applicable to debugging and optimizing web applications.
- Demonstrated capability to work with complex algorithms, aligning with the technical demands of the role.
- Showcased proficiency in Python, a valuable skill for server-side development and integration.
- Applied structured programming techniques, crucial for developing scalable and maintainable codebases.

Automated Personal Budget Management System

Google Apps Script, Google Sheets

- Developed an automated budget management system in Google Sheets to track daily expenses, income, and category-based spending trends.
- Built custom Google Apps Script functions to automate calculations, expense categorisation, and dynamic financial summaries.
- Designed a structured calendar-style layout to visualise daily spending and improve financial planning.
- Implemented automated spreadsheet logic to generate monthly summaries and financial insights with minimal manual input.
- Demonstrated proficiency in automation and scripting, enhancing efficiency in financial data management.
- Showcased ability to create user-friendly interfaces, applicable to developing intuitive web applications.
- Applied problem-solving skills to automate repetitive tasks, aligning with the role's focus on efficiency.
- Enhanced understanding of data visualization techniques, crucial for creating compelling digital marketing materials.

Automated Payslip Generator for Payroll Tracking

Google Apps Script, JavaScript, Google Sheets, Google Docs

- Built an automated payslip generation system that converts payroll data from Google Sheets into structured payslip documents.
- Developed Google Apps Script workflows to dynamically populate employee details, salary calculations, and deductions.
- Automated document generation and formatting to produce consistent, professional payslips directly from spreadsheet data.
- Reduced manual payroll processing by creating a repeatable and scalable document automation pipeline.
- Demonstrated proficiency in JavaScript and scripting, essential for developing automated solutions.
- Showcased ability to design structured workflows, applicable to managing multiple projects simultaneously.
- Applied problem-solving skills to streamline payroll processes, enhancing efficiency and accuracy.
- Enhanced understanding of data management techniques, crucial for developing effective digital marketing materials.

Languages		
English c2 Bilingual or Proficient (C2)	TshiVenda c1 Advanced (C1)	isiZulu c1 Advanced (C1)
isiXhosa B2 Upper-Intermediate (B2)	French A2 Elementary (A2)	

